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Spelling out phases

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Abstract: Evidence from many PF phenomena shows that what is sent to spell-out is phases, not phasal complements. Spelling out phases, however, raises an issue for successive-cyclic movement: if what is sent to spell-out is a full phase and inaccessible to syntax, if successive-cyclic movement were to target phasal edges, as standardly assumed, an element undergoing movement would get frozen with the first step of successive-cyclic movement since it would be part of a spelled-out unit hence inaccessible. The paper shows that the conundrum can be resolved under the labeling theory: spell-out can correspond to a phase and successive-cyclic movement can still target a phase (i.e. phasal edge). The system is shown to resolve a number of issues, e.g. certain stress assignment puzzles in English, tone sandhi issues in Taiwanese, and the seeming C'-deletion with sluicing. It also deduces criterial freezing in a way that captures its parallel with subextraction. The paper also shows that successive-cyclic movement and final movement to the same position can have different reflexes in phonology and leads to a new conception of *edge* in PF.

Keywords: labeling, phases, spell-out, successive-cyclic movement, syntax-phonology interface

1. Introduction

Evidence from a number of phonological/PF phenomena shows that what is sent to spell-out is phases, not phasal complements (see Bošković 2016a and the discussion in this paper), which is in fact what is expected given that only phases have theoretical status, phasal complements do not. Spelling out phases, however, raises an issue for successive-cyclic movement. If what is sent to spell-out is a full phase and inaccessible to syntax, if successive-cyclic movement were to target phases, i.e. phasal edges, as is standardly assumed, an element undergoing movement would get frozen with the first step of successive-cyclic movement since it would be part of a spelled-out unit hence inaccessible. This paper shows that the conundrum can be resolved under the labeling theory: spell-out can correspond to a phase and successive-cyclic movement can still target a phase (i.e. phasal edge). Under the conception of spell-out argued for, spell-out is contextual in the sense that phase XP with YP in its Spec does not always behave in the same way regarding spell-out: sometimes traditional C' is sent to spell-out and sometimes the full CP is sent to spell-out although in both cases the spell-out domain is a full phase. The system will be shown to resolve a number of outstanding puzzles and shed light on a number of issues, e.g. certain stress assignment puzzles in English, seeming C'-deletion with sluicing, criterial freezing... It will also be shown to have consequences for the nature of the syntax-phonology interface, and the notion of *edge* in PF, which will receive a very different treatment from the standard conception of edge.

The paper is organized as follows. Section 2 discusses arguments that what is sent to spell-out is phases, not phasal complements, emphasis being on arguments based on the syntax-phonology interaction. Sections 3 and 4 turn to the main issue this paper addresses, consequences of phasal spell-out for successive-cyclic movement. As noted above, given that what is sent to spell-out is a full phase and that it is inaccessible to syntax, if successive-cyclic movement were to target phases, an element undergoing movement would get frozen with the first step of successive-cyclic movement since it would be part of a spelled-out unit hence inaccessible. Section 4 will present a resolution to this apparent contradiction. The following sections then discuss the consequences of the system proposed in section 4 for a number of phenomena and mechanisms. Section 5 does this for stress assignment in English, section 6 discusses tone sandhi in Taiwanese, and section 7 raddopiamento fonosintactico in Abruzzese. Section 8 discusses the consequences of the proposed system for a number of additional syntactic phenomena, in particular, criterial freezing, subextraction, head-movement, and sluicing. Section 9 discusses more general consequences of the proposed analysis for the nature of the syntax-phonology interface and the notion of edge (in PF). Section 10 concludes.

I will start the discussion by summarizing some of the arguments given in Bošković (2016a) that what is sent to spell-out is phases, not phasal complements. Given the importance of the issue, a number of new arguments will also be provided.[1]

2. What is sent to spell-out is phases, not phasal complements

2.1. Conceptual arguments

2.1.1. The obvious argument

Only phases have theoretical status; theoretically, we would expect that what is sent to spell-out is phases, not phasal complements (this was in fact the earliest assumption regarding phases and multiple spell-out, see Chomsky 2000:131-132, see also Fox and Pesetsky 2005, Bošković 2016a, Cecchetto and Donati 2022, Saito 2024, among others); compare in this respect the great deal of effort that has gone into coming up with the proper definition of *phase* with no attempts at all of that kind regarding phasal complements. The reason is simple: phasal complements have no theoretical status, only phases do.

2.1.2. Matrix clauses

Matrix clauses significantly complicate the phasal complement spell-out approach: if only phasal complements are sent to spell-out, matrix clauses will never be sent to spell-out. Under this approach, we then need an additional assumption regarding matrix clauses to ensure that not just IP but also CP, which is a phase, is sent to spell-out in (1): what is sent to spell-out is phasal complements and matrix clauses.

(1) [$_{\text{CP}}$ What did [$_{\text{IP}}$ John buy]]?

Under the phasal complement spell-out approach, we thus still need to assume that in certain cases what is sent to spell-out is a phase. This is not an innocent assumption given that in many languages, matrix and embedded clauses look exactly the same (cf. English (2)-(3)[2]), which means that at the point when the clause is built, we cannot tell whether we are dealing with an embedded or a matrix clause.

(2) a. John left. b. I think John left.
(3) a. Who left? b. I wonder who left.

The point here is that while phasal complement spell-out needs an additional non-innocent assumption to account for matrix clauses, requiring phasal spell-out in this case, phasal spell-out is consistent in that only phases are ever sent to spell-out.

2.1.3. Labeling and phasal spell-out

Another argument for phasal spell-out comes from Chomsky's (2013) labeling/structure-building system, where unlabeled objects are allowed during the derivation but not in the final representation. As in Collins (2002), labeling is not part of the definition of Merge, which combines two syntactic objects without

[1]In particular, a number of new phonological arguments will be provided. This is important since when the issue in question is discussed in the literature, what is generally addressed is whether spell-out domains are relevant to phonological processes, the issue of whether the spell-out domain corresponds to a phase or a phasal complement is generally ignored. Moreover, the actual empirical arguments in the phonological literature are also often riddled by unwarranted and problematic syntactic assumptions, which, when dissolved, also often dissolve the arguments themselves. It is therefore important to provide a variety of phonological arguments that are as clean as possible when it comes to such assumptions and that crucially tease apart the phase/phasal complement spell-out domain hypotheses.
[2]See Agbayani (2000), An (2007), Bošković (2024), McCloskey (2000), Pesetsky and Torrego (2001), Messick (2020), among others, for arguments that examples like (3a-b) involve wh-movement of *who*.

labeling the newly created object. The traditional assumption that when X and Y are merged, the nature of the resulting object needs to be specified is, however, still kept. Syntax itself does not require this information. It is required by the interfaces so that syntactic objects can be interpreted: there is nothing wrong with unlabeled objects in the syntax, but such objects are uninterpretable. Chomsky then provides an algorithm that specifies labels which applies at the point of transfer to the interfaces (see section 4), since labeling is interface-driven.

Under the interpretation-driven approach to labeling, we would expect labeling to occur at the point of transfer to the interfaces for what is being transferred. However, although Chomsky assumes that what is sent to spell-out is the phasal complement, he crucially assumes that the whole phase is labeled at this point, not just the phasal complement.

There is an inconsistency here: if labeling is interpretation/interface-driven, with syntax itself not requiring labeling, what should be labeled is what is sent to the interfaces. The issue is resolved if what is sent to the interfaces is the whole phase. It is then not surprising that what is labeled is the whole phase, not just the phasal complement.

Chomsky (2013), where labeling is interface-driven and what is being labeled is phases, thus naturally leads to the assumption that full phases, not phasal complements, are sent to spell-out.

In the next section I turn to phonological arguments for phasal spell-out.

2.2. Phonological arguments for phasal spell-out

Under standard assumptions, CP is a phase in (4), and the spell-out domain is IP (its complement). Under phasal complement spell-out, Y should be able to interact with X, but not with Z in PF, while under phasal spell-out, Y should be able to interact with Z, but not X.

(4) X [$_{CP}$ Y [$_{IP}$ Z

Bošković (2016a) provides a number of arguments that Y can interact with Z but not with X. Consider the Bulgarian/Macedonian cliticization in (5) (clitics are given in italics).

(5) a. Vera *mi* *go* dade včera. Bg/Mac:OK
Vera me.dat it.acc gave yesterday
'Vera gave it to me yesterday.'
b. Vera *mi go* včera dade. Bg/Mac: *
c. *Mi go* dade Vera včera. Bg: * Mac: OK
d. Dade *mi go* Vera včera. Bg: OK Mac: *

Bulgarian/Macedonian clitics are verbal clitics, but in Bulgarian they encliticize and in Macedonian they procliticize. Bošković (2001) gives a copy-and-delete account of (5) based on Franks' (1998) (see also Bošković 2001, Bošković and Nunes 2007) proposal that a lower copy of a non-trivial chain can be pronounced in PF iff this is necessary to avoid a PF violation. The clitic undergoes movement "around" the verb. In Macedonian, the higher clitic copy can always be pronounced, hence it must be. In Bulgarian, it can be pronounced when there is something preceding it, but not when nothing precedes it, as in (6b). Here, a lower copy is pronounced since higher copy pronunciation would induce a PF violation. The analysis captures the last resort nature of the V-Cl order in Bulgarian.

(6) a. X cl V ~~cl~~ b. ~~cl~~ V cl (Bg)
c. (X) cl V ~~cl~~ (Mac)

Note that coordination can support Bulgarian enclitics, hence the impossibility of V-Cl order below.

(7) a. Dade *ti* *go*.
gave you.dat it.acc
'(S)he gave it to you.
b. I *ti* *go* dade.
and you.dat it.acc gave
c. *I dade *ti* *go*. (Bg)

Consider now Bulgarian/Macedonian questions with the interrogative C *li*, also an enclitic. They involve V-to-C movement, with the V carrying the clitics along, as in Macedonian (8a). Surprisingly, in Bulgarian (8b-c), the conjunction apparently cannot support the enclitics.

(8) a. I *ti* *go* dade *li*?
and you.dat it.acc gave Q
'And did (sh)he give it to you?' (Mac)
b. I dade *li* *ti* *go*?
and gave Q you.dat it.acc
'And did (s)he give it to you?' (Bg)
c. *I *ti go* dade *li*? (Bg)

Given that the conjunction can satisfy the enclitic requirement (see (7)), (8c) should be acceptable. Instead, what we get is (8b). Why does lower copy pronunciation apply in (8b), although it does not in (7a)?

Franks and Bošković (2001) provide a multiple spell-out account, where what is sent to spell-out is crucially the phase. The cl+V head cluster moves to *li.* The CP is then sent to spell-out, with a lower copy of the clitics pronounced since there is nothing in (9) that can support the enclitics.

(9) [$_{CP}$ [$_{C}$ ~~ti go~~ dade+li] ti go ~~dade~~]

The conjunction *i* is only then added and we derive (8b):

(10) i [$_{CP}$ [$_{C}$ ~~ti go~~ dade+li] ti go ~~dade~~]

Now, if phasal complements were sent to spell-out, C would be in the same spell-out domain as the conjunction. This should then lead to higher copy pronunciation in Bulgarian, just as in Macedonian (see (11)). This apparently does not happen in Bulgarian, which means the whole CP phase is sent to spell-out.

(11) a. [$_{CP}$ [[ti go dade] li] ~~mi go dade~~]
b. i [$_{CP}$ [[ti go dade] li] ~~mi go dade~~]

Consider now stress assignment in all-new neutral sentences in German, within the approach to phase-based pitch accent assignment from Kratzer and Selkirk (2007). The relevant assumptions from Kratzer and Selkirk (2007) are given below, slightly modified (see Kahnemuyipour 2003 for an early version of (13)):

(12) The spell-out domain is the prosodic domain for phrase stress.
(13) Phrase stress is assigned within the highest <u>phrase</u> within the spell-out domain.
(14) The Elsewhere Condition on Prosodic Spell-out: Phrase stress must be assigned within a spell-out domain with eligible material.

To see how this works, consider (15), assuming an all-new neutral context. Kratzer and Selkirk argue that the bracketed element corresponds to a spell-out domain. The highest phrase within the domain, *eine Géige,* receives phrase stress.

(15) … dass ein Júnge [eine Géige an einen Freund schickte].
that a boy a violin to a.ACC friend sent
'…that a boy sent a violin to a friend.' (Kratzer and Selkirk 2007:107)

To determine whether the spell-out domain corresponds to a full phase or a phasal complement, consider (16), where both the subject and the object of the embedded clause receive phrase stress.

(16) Ich glaube, dass María die Gesétze studiert.
I think that Maria the laws is.studying
'I think that Maria is studying the laws.' (Kratzer and Selkirk 2007:94)

The line of research going back to Kayne (1994) and Zwart (1993) argues that the OV order in German results from object shift, i.e. movement to SpecvP. Even independently of this line of research, Diesing (1996) argues that non-contrastive definite objects must move in German. In other words, the object in (16) is generally assumed to undergo object shift/scrambling.

Given antilocality (the ban on movement that is too short), which bans movement within a phrase (see Bošković 1994, 2013a, Saito and Murasugi 1999, Abels 2003, Grohmann 2003, among many others), this movement can target SpecvP or a higher position, i.e. SpecvP is the lowest potential landing site. Importantly, for the object to be stressed it cannot be located in the same spell-out domain as the subject. If the object moves to SpecvP, and what is sent to spell-out is the whole vP phase, the object and the subject will be in different spell-out domains. Each element will then receive stress. On the other hand, if what is sent to spell-out is the phasal complement, which is VP, the object and the subject would be in the same spell-out domain, which means the object could not be stressed. The phasal spell-out system thus captures the stress pattern in (15), where both the subject and the object bear phrase stress, which is problematic for the phasal complement spell-out approach.

Consider also the contrast in (17)-(18).

(17) Ich hab' gelesen, dass die Metállarbeiter gestreikt haben.
I have read that the metal.workers gone.on.strike have
'I read that the metal workers went on strike.' (Kratzer and Selkirk 2007:95)

(18) Ich hab' gehört, dass der Rhéin stínkt.
I have heard that the Rhine stinks
'I've heard that the Rhine stínks.' (Kratzer and Selkirk 2007:96)

(17) involves a stage-level, and (18) an individual-level predicate. Diesing (1992) argues that the subject in (17) can be located either in SpecvP or SpecTP in overt syntax, while the subject in (18) must be in SpecTP. Importantly, the verb can be unstressed only in (17). Under phasal spell-out, a derivation is available for (17) on which the subject and the verb are in the same spell-out domain, the subject staying in SpecvP. The subject then receives phrase stress as the highest phrase within the vP spell-out domain, which corresponds to the vP phase. This is not an option in (18), where the subject cannot be in SpecvP. The subject being outside of vP in (18), the verb in v is the only element in the vP phase, which is sent to spell-out, hence it receives stress under (14). ((14) assigns stress to the verb, which is a head, not a phrase, when there is nothing else in the spell-out domain that can be stressed.)[3] What is important is that the verb must be

[3]Since the subject of stage-level predicates can move outside of vP (Diesing 1992), the phasal spell-out account predicts that the verb in (17) can be optionally stressed, which is indeed the case, as Kratzer and Selkirk (2007) note.

stressed in (18) under phasal spell-out. This is, however, not the case under phasal complement spell-out. The verb, located in v, does not get stress in (18) since it belongs to the same spell-out domain as the subject in spite of the subject being outside vP, if what is sent to spell-out is VP. The verb in fact cannot get stress in either (18) or (17) (see also fn 3): regardless of whether the subject moves to SpecTP it is in the same spell-out domain as the verb. Stress assignment in all-new neutral sentences in German thus favors phasal spell-out.

Consider now the interaction of scope and prosody in Japanese, based on the data below:

(19) zen'in-ga sono tesuto-o uke-nakat-ta. (SOV)
all-NOM that test-ACC take-NEG-PST
'All did not take that test.' *not >> all, all >> not (Ishihara 2007:139)

(20) sono tesuto-o$_i$ zen'in-ga t$_i$ uke-nakat-ta. (OSV)
that test-ACC all-NOM take-NEG-PST
'That test, all didn't take.' not >> all, all >> not (Miyagawa 2003:183-184)

Miyagawa (2003) argues that the subject located in SpecTP must scope over the negation, while the negation obligatorily scopes over the subject in SpecvP. The subject in (19) moves to SpecTP, hence it scopes over the negation. Miywaga argues that (20) is ambiguous structurally: on one derivation, the subject moves to SpecTP and the object moves to SpecCP—the subject then takes wide scope. When the negation takes wide scope, the object undergoes A-movement to SpecTP, which satisfies the EPP, with the subject staying in SpecvP. Interestingly, Ishihara (2007) notes that prosody disambiguates (20). When there is a major phrase boundary after the subject, the subject must take wide scope (21). However, under the prosodic phrasing in (22), the subject cannot take wide scope. Moreover, the unambiguous (19), where the subject moves to SpecTP, patterns with (21) in this respect (see (23)).

(21) (all >> not) [$_{CP}$ sono tesuto-o$_i$ [$_{TP}$ zen'in-ga$_j$ [$_{vP}$ t$_i$ t$_j$ [$_{VP}$ t$_i$ uke-nakat-ta]]]]
(sono tesuto-o)$_{MaP}$ (zen'in-ga)$_{MaP}$ (uke-nakat-ta)$_{MaP}$

(22) (not >> all) [$_{TP}$ sono tesuto-o$_i$ [$_{vP}$ t$_i$ zen'in-ga [$_{VP}$ t$_i$ uke-nakat-ta]]]
(sono tesuto-o)$_{MaP}$ (zen'in-ga uke-nakat-ta)$_{MaP}$

(23) (all >> not) [$_{TP}$ zen'in-ga$_i$ [$_{vP}$ t$_i$ sono tesuto-o uke-nakat-ta]]
(zen'in-ga)$_{MaP}$ (sono tesuto-o uke-nakat-ta)$_{MaP}$ (Ishihara 2007:147)

Ishihara (2007) proposes that major phrases correspond to spell-out domains. However, spell-out domains crucially have to correspond to full phases, not phasal complements. As a result, vP, not its VP complement, is what is sent to spell-out, hence the major phrase boundary at the edge of vP in (21)-(23).

The above arguments for phase spell-out come from the impossibility of interaction between the edge of phase XP and material outside XP: this is predicted under phase spell-out since the two belong to different spell-out domains but not under phasal complement spell-out, where they are in the same spell-out domain.

(24) **K** [$_{XP}$ **YP X** [$_{ZP}$ L XP is a phase

Also relevant is PF interaction between a phase head and its complement: under phasal spell-out, such interaction should be possible since the two belong to the same spell-out domain. It should be impossible under phasal complement spell-out since the two do not belong to the same spell-out domain. Bošković (2016a) gives a number of arguments that such interaction is possible. Consider raddoppiamento fonosintattico (RF) in Abruzzese. In RF, the initial consonant of a word undergoes gemination which is conditioned by the properties of the preceding word. A lexically conditioned set of words triggers RF when they are in a specific syntactic relationship with the following word. The complementizer *chə* is an RF trigger. In (25), it triggers RF on the first word in its IP complement. In (26), RF applies between the relative pronoun in SpecCP and the subject. Crucially, based on active/passive contrasts regarding RF, D'Alessandro and Scheer (2015) and Bošković (2016a) argue that RF is spell-out domain sensitive.[4] We then have here an argument that phasal complements are not spell-out domains. (Another argument to this effect based on tone sandhi in Taiwanese will be given in section 6.)[5]

(25) Jè mmeje **chə vve**.
is better that come.3SG
'It's better that he/she comes.'

(26) lu waglionə **chə ttu** si vistə
the boy whom you are seen
'the boy whom you saw' (D'Alessandro and Scheer 2015)

Additional evidence for phasal spell-out not noted in Bošković (2016a) comes from Malayalam causatives.[6] What is important here is that when unaccusatives are causativized, phonological processes can apply across the verb-suffix boundary. Thus, in (27) the final consonant of the root coalesces with the causative suffix *ikk*. In (27b), the [ai] vowel hiatus is resolved by deleting the initial [i] of the suffix.

(27) Causative attached to unaccusatives
a. /aaʈ+ikk/ = [aaʈʈ] `Y shakes X'
shake$_{root}$+cause
b. /nana+ikk/ =[nanakk] `Y waters X'
water$_{root}$+cause (Kilbourn-Ceron et al. 2016:126)

Importantly, when unergatives are causativized, neither coalescence nor vowel deletion is triggered. Thus, in (28a), the final consonant of the verb does not coalesce with the causative suffix and in (28b), no vowel deletion occurs: instead, hiatus is resolved by inserting the glide [j] between the root and the suffix.

(28) Causative attached to unergatives
a. /paaʈ+ikk/ [paaʈikk] 'Y makes X sing'
sing$_{root}$+cause
b. /kaɹa+ikk/ [kaɹajikk] 'Y makes X cry'
cry$_{root}$+cause (Kilbourn-Ceron et al. 2016:126)

[4]The active/passive contrast is illustrated by (i) (see Bošković 2016a for an account, where the auxiliary is outside of the vP phase in (i)).

(i) So rəspəttatə la leggə.
am.1SG respected.SG the.F.SG law.F.SG
'I have respected the law.' (D'Alessandro and Scheer 2015:611)

(ii) **So rrəspəttatə** (da tuttə quində).
am.1SG respected.SG by all
'I am respected by everybody.' (D'Alessandro and Scheer 2015:612)

[5]For an additional argument of this sort regarding phonological interaction between complementizers and subject clitics in Arabic, see Bošković (2016a)..

[6]The Malayalam data below are taken from Kilbourn-Cerone et al (2016), who argue based on these data that cyclic phonological processes are phase-sensitive but do not address the narrower question of what is sent to spell-out, phase or phasal complement, which we are interested in here (the analysis proposed below also differs from their analysis; note that the remarks from this footnote also apply to the discussion of Moro below.

What the above data then show is that phonological processes can apply between the verb and the causative suffix with unaccusatives (29), but not between the verb and the causative suffix with unergatives (30). I assume that in both (27) and (28), the causative suffix is attached to vP. Following Chomsky (2000), with unergatives, this vP is a phase, and with unaccusatives, it is not. (28) then shows that phonological processes are not possible across a phasal boundary, even if the relevant element is located at the phase edge. The problem does not arise in (27), since vP is not a phase here. The above data thus not only provide evidence that phonological domains coincide with spell-out domains, as Kilbourn-Ceron et al. (2016) note, but they further show that the spell-out domain corresponds to a phase, not a phasal complement. If it were to correspond to the phasal complement, v in (29) should be able to phonologically interact with the causative affix, since it is located at the phasal edge.

(29) [$_{vP}$ Verb] -ikk] unergatives: vP a phase
|__*__|

(30) [$_{vP}$ Verb] -ikk]] unaccusatives: vP not a phase
|___|

Also relevant are certain data from Moro discussed by Sande et al (2020). In Moro, certain inflectional categories trigger a left-aligned high tone on the stem. In (31a), this tone is realized on the leftmost syllable of imperfective stems. For perfective stems, as in (31b), this tone is not realized. (31c) shows that the absence of an H tone is not due the presence of a final H tone on the suffix.

(31) Moro verbs (Sande et al. 2020: 1212)

	Verbal Inflection	PreV-[‘tickle’-*v*]	Gloss
a.	Imperfective	ga-[tʃómbəð-a]	‘S/he is tickling it’
b.	Perfective	ga-[tʃombəð-ó]	‘S/he tickled it’
c.	Venitive Imperative	[tʃombəð-a]	‘Tickle it and come’

Importantly, elements in the preverb field (i.e. tense/aspect/mood/agreement morphology; see Rose 2013 and Jenks and Rose 2015 for discussion of Moro preverbal field morphology—I will simply refer to it as Infl) do not participate in this H tone assignment. As (31a) shows, no H tone is placed on *ga-* despite *ga-* being the leftmost syllable of the *ga*+stem word. Sande et al (2020) interpret the observation that tone changes may only operate on the root but not the preverb as indicating that H tone assignment is phase/spell-out domain sensitive, but do not discuss the narrower question that we are interested in. Let us see then see how these data can be accounted for. Following Sande et al (2020), imperfectives and perfectives have a distinct v (see this work for evidence to this effect). Further, the final vowel on the verb stem is the realization of the phase head *v*, with $v_{[ipfv]}$ triggering H tone assignment (see Sande et al 2020 for details), which $v_{[pfv]}$ does not do. I further assume that, as discussed above, spell-out domains correspond to phases, not phasal complements. Then, in (31a), when the vP, i.e. root+$v_{[ipfv]}$, is spelled out, an H tone is assigned to the leftmost syllable of the stem. Infl (i.e. preverb ga-) is then merged into the structure. In the next spell-out domain where Infl is sent to spell out, since Infl (i.e. *ga-*) is outside of the vP phase in which the tone-shift triggering $v_{[ipfv]}$, the head of the vP phase, was spelled out, no tone is assigned to the preverb, as in (31a). Importantly, the data in (31a) indicate that *v* must be spelled out in the same cycle as the verb stem, but crucially not with Infl. If $v_{[ipfv]}$ were spelled-out in a higher phase, with Infl, i.e. the preverb *ga-*, H tone should be assigned to *ga-* in (31a) since it would be the leftmost syllable in the H tone assignment domain. This in turn indicates that the spell-out domain is the full vP phase, not the VP complement of v.

Another argument that the phase head is spelled out together with its complement is provided by certain Mongolian data discussed by Barrie and Kang (2022). They propose that in (32), when vP is sent to spell-out, an epenthetic vowel is inserted because [wl] is not a well-formed coda in Mongolian. This vowel remains after T is added on the next cycle. (The important part of the form in question is given in bold below. Note that I have changed the underlying forms given by Barrie and Kang to reflect applications of a vowel harmony rule, not discussed in that work (for relevant discusson, see Svantesson et al 2005). I have also slightly adjusted the structures given by Barrie and Kang).

(32) cowl-lo [$_{TP}$ [$_{vP}$ **co.wel**.] lo]
advise-PST
'advised' (Barrie and Kang 2022: 413)

Consider, however, (33):

(33) cowl-l-o [$_{nP}$ **cow.le.]** lo]
advise-NZLR-REFL
'his advice' (Barrie and Kang 2022: 414)

(33) involves the nominalizer [l]. Barrie and Kang argue that nP is a phase, with the nominalizer being the phase head. When the nP phase is sent to spell-out and the *n* head, realized as [l], attaches to the root [cowl], we get [cowll]. An epenthetic vowel is then inserted to break up the illegitimate code. Importantly, since *n* and the root are spelled out together, the epenthetic [e] is inserted between the [ll] sequence, so that the form is realized as [cowlel] (compare this to [cowel] in (32)). In the next cycle, when the reflexive head, i.e. [o], is added, the coda consonant of the root+n complex is resyllabified as the onset of the final syllable. What is important here is that if the root in (33) were spelled out in a separate cycle from *n,* the epenthetic vowel would be inserted in [wl], as in (32). But instead, as (33) shows, the epenthetic vowel is inserted in [ll]. This shows that the phase head *n* and the root are spelled-out in the same domain, thus, they are in the same domain for vowel epenthesis.[7]

The above phenomena illustrate the impossibility of PF interaction between the edge of phase XP and material outside of XP, as well as the possibility of PF interaction between the edge of phase XP and the complement of X. The conclusion is then that spell-out domains correspond to phases.

It should also be noted that pre-phasal approaches to prosodic phrasing (e.g. Nespor and Vogel 1986, Selkirk 1986) anticipated phasal spell-out since the standard assumption in these approaches is that the CP edge corresponds to an intonational-phrase boundary (in fact, this is still the standard assumption)—the correspondence is with a phase, not a phasal complement, which is natural if spell-out domains correspond to phases. On the other hand, we have a rather strange situation here under phasal complement spell-out: under this approach, what is sent to spell-out is the IP below CP and the VP above the CP, but the prosodic correspondence is with the "sandwiched" phrase, CP.

[7]To briefly mention here another new argument (the relevant data are too complex/long to present here), in Bošković's (2013a) approach to phases, where every lexical category projects a phase, with the highest phrase in the domain of a lexical category being a phase, adjectives also project a phasal domain. Talić (2015, 2023) provides syntactic and prosoding evidence for the presence of a functional head X (see these works for its precise nature), which is the highest projection in the adjectival domain, making XP a phase (see Talić 2015, 2023 for evidence for its phasehood). Crucially, Talić shows that X interacts with what is in its complement with respect to accent assignment, whose domain she shows corresponds to a spell-out domain. While this requires additional assumptions in the phasal-complement spell-out model (see Talić 2023), it follows straightforwardly from the phase spell-out model.

3. Back to successive-cyclic movement

But what about successive-cyclic movement (SCM) if what is sent to spell-out is phases, not phasal complements? If both multiple spell-out and SCM were to be defined on phases, phases would be spell-out units and SCM would target phases. A problem then arises: if what is sent to spell-out is no longer accessible to the syntax, the element undergoing SCM would get frozen with the first step of SCM since it would be part of a spelled-out unit, hence no longer accessible. It appears then that either spell-out or SCM can be stated in terms of phases, but not both. Above, we have seen a number of arguments that what is sent to spell-out is phases. Adopting this assumption, Bošković (2016a) proposes an approach to SCM where what is sent to spell-out is indeed a phase. Following Chomsky's (2001) assumption that what is targetted by SCM is the first phrase above the phrase that is sent to spell-out, Bošković (2016a) argues that SCM does not target phasal edges, but the edge of the first phrase above the phase.

Here is what we seem to need: In (34), where XP is the first phrase above YP and these phrases are affected by SCM and spell-out, YP should be the spell-out domain, and XP should be targeted by SCM, with YP spelled out after that movement. Furthermore, only one of these should correspond to a phase.

(34) [$_{XP}$ [$_{YP}$

The question is which of the two, XP or YP, should be a phase; i.e. the issue is whether the domain for SCM or the domain for spell-out should correspond to phases. For Chomsky (2001), XP is the phase: SCM domain is defined on phases, spell-out domain is not. In principle, it can be the other way round, with YP being a phase, not XP. Spell-out domain would then be defined on phases, not SCM domain. This is what Bošković (2016a) proposes. Chomsky's (2001) approach is based on phasal complement spell-out, while Bošković's is based on phasal spell-out. They are illustrated in (35)-(36), where what is bolded is a phase and what is shaded is the spell-out domain (they are the same for Bošković 2016a).

(35) What do you think **[$_{CP}$** what that [$_{IP}$ Mary bought]] (Chomsky 2001)
(36) What do you think [$_{VP}$ what **[$_{CP}$** that Mary bought]] (Bošković 2016a)

Recall that given that what is sent to spell-out is no longer accessible to the syntax, it is not possible to state the domain for both spell-out and SCM in terms of phases. If SCM were to target spell-out units, the moving element would get trapped since it would be part of a spelled-out unit. Only one of the two, spell-out or SCM, can then be stated in terms of phases. This is the property of both Chomsky (2001) and Bošković (2016a). In both, XP is sent to spell-out and movement targets YP right above it (with XP spelled out after that movement). In both, one of the two is defined on phases. The difference is which mechanism is defined on phases. For Chomsky, it is SCM—SCM targets phases, spell-out doesn't. In Bošković (2016a), spell-out targets phases, SCM doesn't. We are dealing here with the issue of primacy: what should be privileged, spell-out or SCM. By defining the former on phases, with SCM piggy-backing on it, Bošković (2016a) privileges spell-out, while Chomsky privileges SCM. Many have argued that SCM takes place so that the moving element escapes being sent to spell-out (e.g. Bošković 2007, Fox and Pesetsky 2005, Stjepanović and Takahashi 2001). This appears to favor a system where spell-out is privileged, i.e. where spell-out is defined on phases, but SCM is not.

However, in the following section I will show that the conundrum that we have been dealing with can be resolved under Chomsky's (2013) labeling theory.[8] In particular, it will be shown that under the labeling

[8]For relevant discussion under different assumptions regarding labeling and structure building, see Cecchetto and Donati (2022).

theory, spell-out can correspond to a phase and SCM can still target a phase (i.e. traditional phasal edge). This is conceptually appealing: under the phase theory we want everything, i.e. all domains, to be defined on phases. That the labeling theory can achieve a significant unification here, unifying spell-out and SCM in that both target phases, should then also be interpreted as an argument for the labeling theory.

4. Resolving the spell-out/successive-cyclic-movement conundrum: Phases and labeling

Chomsky (2013) provides a labeling algorithm where when a head and a phrase merge, the head projects (i.e. it provides the label for the resulting object). When two non-minimal projections (phrases) merge, there are two ways of implementing projection/labeling: prominent feature sharing or traces (movement), traces being ignored for labeling.

(37) illustrates the former: when *what* merges with interrogative CP in (37), both *what* and the CP have the Q-feature, which determines the label of the resulting object.

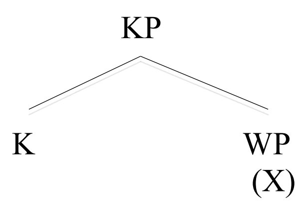
(37) I wonder [$_{CP}$ what$_i$ [$_{C'}$ C [John bought t$_i$]]]

Consider now SCM in (38). SCM does not involve feature sharing (which follows Bošković 1997, 2002, 2007, 2008): there is no feature sharing between *that* and the wh-phrase passing through its edge. Labeling via feature sharing not being an option, the embedded clause cannot be labeled when *what* moves to its edge (39). After v merges, *what* moves. The element merged with *that*-CP being a trace, it is ignored for labeling, hence ? is labeled as CP after *what* moves (39). Only at this point the status of t'$_i$ in (38) can be determined as SpecCP. Prior to the movement (39), ? is not a CP, its status is simply undetermined.

(38) What$_i$ do you think [$_{CP}$ t'$_i$ [$_{C'}$ that [John bought t$_i$]]]
(39) v [$_{VP}$ think [$_?$ what [$_{CP}$ that [John bought t$_i$]]]

In addition to the labeling theory, the account of SCM to be proposed will need another ingredient, which is the way phase determination is implemented in contextual approaches to phases. In particular, Bošković (2012, 2013a, 2014) argues for a contextual approach to phases where there are phasal domains and the highest phrase in a phasal domain is a phase.[9] In this system, phase is determined when a phasal domain is closed. This is in turn determined when the element with the relevant phasal property (X) no longer projects, which closes the phasal domain—X determines a phase when X no longer projects. WP in (40) then becomes a phase only after WP merges with K, with K projecting (i.e., with WP no longer projecting).[10]

(40)



In other words, the phasal status of WP is determined only after it merges with K (with WP not projecting). Note here the importance of "further merger".

Let us now return to SCM and phasal spell-out.

[9]This approach gains in theoretical significance with the contextual approach to the EPP proposed in Bošković (2024), where there is an EPP domain, with the highest phrase in the EPP domain functionining as the locus of the EPP, as well as the approach to phasal edges in Bošković (2016c), where in cases of multiple edges of the same phase, only the highest edge counts as the edge.
[10]Note that if K had X, WP would not be a phase.

(41) What do you think that John bought?
(42) [$_?$ what **[CP** that [$_{TP}$ John bought]]]

In (42), C and TP merge, with the resulting object labeled CP. *What* then moves, merging with the CP. The resulting object is unlabeled. Since there was a further merger and CP didn't project, the phase domain is closed. As the highest projection in the phasal domain, the CP sister of the wh-phrase is a phase and sent to spell-out, given that what is sent to spell-out is phases. (Phase/spell-out domain is given in bold in (42)/(44)). The immediate daughters of ? are *what* and the CP. Once *what* moves away (after merger of additional structure), traces being ignored for labeling, ? is labeled as CP.

We can then have spell-out correspond to a phase and SCM target a phase. *What* targets CP (*that*+IP), and this CP is sent to spell-out. *What* is outside of the spell-out domain, but not because of the PIC, which is actually not needed, but because it is not part of the phase that is sent to spell-out.

Consider now the case of agreeing Specs.

(43) Mary wonders what John bought
(44) **[CP** what [$_{CP}$ +wh-C [$_{TP}$ John bought]]

Here, there is feature-sharing after *what* and CP merge. The resulting object is determined to be a phase after merger with V, and sent to spell-out. (Phase/spell-out domain is given in bold.)

Traditionally, *what* in (41) and (43) is in the same position, and at the phase edge (i.e. outside the spell-out domain), but in the above analysis these constructions differ regarding whether *what* is in the spell-out domain. What is particularly important is that phase XP with YP in its Spec does not always behave in the same way for spell-out, which indicates contextuality of spell-out: With SCM, the sister of YP is sent to spell-out, otherwise the full XP is sent to spell-out, although both cases involve phasal spell-out.[11] At any rate, the main point of this section is that the resolution to the conundrum that phasal spell-out, which has independent motivation, raises for successive-cyclic movement (namely, given that what is sent to spell-out is a full phase, if successive-cyclic movement were to target phases, an element undergoing movement would get frozen since it would be part of a spelled-out unit hence inaccessible) falls out for free from independently motivated proposals regarding labeling and phases, which provide us with a unified theory of spell-out domains and successive-cyclic movement where both are defined of phases, i.e. where both spell-out and SCM target phases.

In the following section I will apply the above system to several actual cases. What will be important from the discussion below is not just that the system can account for otherwise puzzling data, but that it predicts that successive-cyclic movement and final movement to the same position can have different reflexes in phonology.

5. Stress assignment in English

I will first consider stress assignment in English. In (45)-(47), the word bearing primary stress is underlined.

(45) a. Mary fixed the <u>bike</u>.
b. Mary <u>fixed</u> it. (Legate 2003:511)
(46) a. Mary liked the proposal that George <u>leave</u>.

[11]This contextuality of spell-out gains theoretical significance within a broader theoretical move toward contextuality in syntax which was demonstrated in Bošković (in press b) to hold in a number of domains, namely structure building, labeling, islandhood, phases, phasal edges, and the EPP (see also fn 9).

b. Mary liked the proposal that George left. (Bresnan 1971)

(47) a. Please put away the dishes.

b. ?Please put the dishes away. (Legate 2003:512)

Since the discussion in this section will rely on several assumptions from Legate (2003), I will simply adopt the approach to stress assignment in English adopted by Legate. Legate assumes the Nuclear Stress Rule (NSR), where the primary stress in English is assigned to the final stress bearing element in the VP (i.e. the rightmost element), hence the contrast in (45) (*it* cannot be stressed). The NSR is also responsible for stress assignment in (46a). As for (46b), which is the original argument for multiple spell-out from Bresnan (1971), the NSR applies cyclically: the NSR applies before *proposal* moves from the most embedded object position (Vergnaud 1974, Kayne 1994), assigning stress to *proposal.* In (47a), the NSR applies normally, assigning stress to *dishes*. Importantly, in (47b), which is generally assumed to involve object shift of *the dishes* (see e.g. Johnson 1991, Lasnik 1999, Gallego and Uriagereka 2007), the NSR assigns stress to *away*. Apparently, in (46b) the NSR applies before the object moves, assigning stress to it, but in (47b) it applies after the object moves, hence it does not assign stress to it. Why the difference?

Under standard assumptions regarding these constructions, phases, successive-cyclic movement, and spell-out (where phasal complements are sent to spell-out), *the dishes* in (47b) undergoes object shift to SpecvP. Since vP is a phase, the object in (46b) also moves to SpecvP. After object movement, the VP complement of v is sent to spell-out in both (46b) and (47b). (46b) and (47b) then have the same derivation in all relevant respects (see (48), which gives the relevant part of the structure): t_i is sentence final, hence *the dishes/proposal* is in the position to be assigned stress by the NSR before the movement.

(48) [$_{vP}$ the dishes/proposal$_i$ [$_{VP}$ …t_i]]

While it is difficult to differentiate (46b) and (47b) with respect to stress assignment under the above analysis, which relies on standard assumptions regarding phases, SCM, and spell-out, the contrast is in fact exactly what is expected under the system argued for in the previous section.

The crucial difference between (46b) and (47b) is that (46b) is a case of SCM, which means there is no labeling after the relevant merger, on a par with (41)-(42). Like the CP sister of the moved element in (41)-(42), the vP sister of the moved element in (46b) is sent to spell-out.

(47b), on the other hand, parallels (43)/(44), which means that the moved element is part of the structure that is sent to spell-out.

The objects actually move to the same position in (46b) and (47b), but in (46b) the object is not part of the vP phasal spell-out domain and in (47b) it is.

(49) [$_?$ the proposal$_i$ **[**$_{\mathbf{vP}}$ [$_{VP}$ …t_i]]] (46b)

(50) **[**$_{\mathbf{vP}}$ the dishes$_i$ [$_{VP}$ …t_i]] (47b)

The object thus moves to a position outside the lowest spell-out domain in (46b), while in (47b) it moves to a position within the lowest spell-out domain. This means that there is only one copy of the object in the input to PF in the first phase of (46b), but two copies in (47b).

In (47b)/(50), the PF operation that deletes non-initial copies within a spell-out domain applies, deleting the lower copy of the object. The NSR then assigns stress to *away*, the rightmost element in the spell-out unit.

On the other hand, in (46b)/(49), since there is only one copy of the object in the first spell-out domain, the PF deletion operation that deletes non-initial copies, which treats each phase as a separate unit, does not delete it when it applies to the first spell-out domain. The NSR applies, assigning stress to *proposal*, as the rightmost element within this spell-out domain. (At a later spell-out domain, this occurrence of *proposal* is deleted in favor of a higher occurrence, with the primary stress realized on this higher occurrence i.e. the occurrence that is not deleted, see Legate 2003).

What is important and what differentiates (46b) and (47b) is that the object in (46b) moves outside of the lowest spell-out domain, while in (47b) it moves within it. There is then only one copy of the object in the first spell-out domain of (46b), but two in (47b).

There is a potential interferring factor regarding the Lebeaux effect, illustrated by (51). In contrast to (51a), (51b) does not display a Condition C effect. Lebeaux (1988) accounts for this contrast by arguing that adjuncts can be inserted into the structure acyclically. The relative clause in (51b) is then introduced into the structure acyclically after wh-movement, hence the pronoun never c-commands the R expression on this derivation. This opens up the possibility that the relative clause in (51c) is added acyclically after stress assignment (via the NSR).

(51) a. *Which argument that John$_i$ was wrong]$_j$ did he$_{\cancel{i}}$ accept t_j in the end?
b. [Which argument that John$_i$ had criticized]$_j$ did he$_i$ accept t_j in the end?
c. Mary liked the <u>proposal </u>that George left.

In fact, an account along these lines was proposed by Ochi (1999) to explain why adverbs do not block affix hopping (Chomsky 1957). Under affix hopping, the past tense affix hops onto the verb stem in (52). Since affix hopping requires adjacency it is blocked in (53), which triggers *do*-insertion. To account for (54), it is standardly assumed (e.g. Bobaljik 1995) that adjuncts do not count for adjacency that is required for affix hopping.

(52) a. John ed want it b. John wanted it
(53) a. John ed not want b. John didn't want it
(54) John always wanted it

Ochi (1999) deduces this effect from multiple spell-out and Lebeaux's acyclic adjunction. He proposes that *always* in (55) is inserted acyclically (in the syntax) after the structure is sent to PF for affix hopping.

(55) Mary always goes there: *Mary pres go there* sent to PF, then *always* inserted

Following this reasoning, when the vP phase is sent to spell-out in (51c), the relative clause would not be present. The relevant spell-out domain would have *proposal* as the final stress-bearing element, which means that the NSR would then assign it stress. The relative clause would then be acylically added. This is a very different derivation from the one discussed above. Crucially, it does not involve SCM via SpecvP. Notice, however, that this alternative acyclic-adjunct-insertion derivation is not an option in (56), where *books* bears primary stress, hence the point made above regarding (51c) can be made regarding (56) without the potentially interfering factor noted above.

(56) John asked what <u>books</u> Helen had written. (Bresnan 1971)

6. Tone sandhi in Taiwanese

I turn now to tone sandhi in Taiwanese. In tone sandhi, the lexically listed citation form of a syllable undergoes a rule-governed modification when preceding another tone-bearing syllable in the same tone sandhi domain. In (57), tone sandhi, indicated by •, applies between the Comp *kong* and its IP complement, with *kong* undergoing a tone sandhi change.[12]

(57) A•hui liau•chun• <u>kong</u>• A•sin si• tai•pak• lang.
Ahui thought C Asin is Taipei person
'A-hui thought that A-sin is from Taipei.' (Simpson and Wu 2002:79)

The IP complement can also precede *kong*. In such cases (see the bracketed IP in (58)-(59)), tone sandhi still applies between *kong* and the IP complement (more precisely, what used to be its IP complement). This is surprising since final elements do not undergo tone sandhi—thus, *ho* cannot undergo it in (60) (cf. the lack of a dot following *ho*).

(58) [A•hui liau•chun• kong• A•sin si• tai•pak• lang] <u>kong</u>•.
A-hui thought KONG A-sin is Taipei person KONG
'A-hui thought that A-sin is from Taipei.' (Simpson and Wu 2002:80)

(59) [A•-sin si• tai•pak• lang] <u>kong</u>•.
A-sin be Taipei person KONG
'A-sin is from Taipei.' (Simpson and Wu 2002:81)

(60) A•-sin chin• ho. A•-hui ma• chin• <u>ho</u>.
A-sin very good A-hui also very fine
'A-sin is very well. A-hui is also very well.' (Simpson and Wu 2002:73)

Simpson and Wu (2002) give an argument for multiple spell-out based on these data that involves IP-movement, more precisely, where in *kong*-final cases the IP moves, leaving the complementizer sentence-final.[13] Given that sentence-final elements do not participate in tone sandhi, tone sandhi in (58)-(59) cannot take place after the IP undergoes movement. Rather, it must take place before the movement, when the structure is similar to what we see in (57). This means that tone sandhi is licensed derivationally, on a par with stress assignment in (46b), which can also be implemented through multiple spell-out (see below).

A question then arises, why is there no derivational tone sandhi licensing in (61), i.e. why doesn't the sentence final verb in (61) undergo tone sandhi? Taiwanese is an SVO language, which means that the object in (61) undergoes movement. Prior to the movement, it was in the right position for tone sandhi licensing. Why wasn't tone sandhi licensed on *kong* derivationally in (61), on a par with (58)-(59)?

(61) A•-sin [tai•oan•oe] <u>kong</u>.
A-sin Taiwanese speak
'Taiwanese, A-sin speaks.' (Bošković 2016b)

To answer this question, consider examples like (58)-(59) in more detail. Note first that the complementizer *kong* is a grammaticalized form of the verb meaning 'say' (see in fact (61)). Based on this, Bošković (2016b) argues that there is a null 'say' in (58)-(59), as shown in (62) (there are in fact semantic effects associated with it; there is an additional "I am telling you" meaning present, which Bošković 2016b argues is the result

[12]There are rather complicated conditions for tone sandhi changes that I cannot go into here, and which do not affect the main point of this section (for relevant discussion, see Simpson and Wu 2002 and references therein). So that the reader can focus on what is relevant to our discussion, the relevant element that undergoes tone sandhi, or that is intended to undergo it, is underlined in the examples below.

[13]Given antilocality, which prevents Complement-to-Spec movement within a single phrase, we may actually be dealing here with a Split CP.

of the presence of null *say*). The IP moves into the domain of this null *say*.[14] Since CP is a phase, the movement proceeds through the edge of CP (62). However, since we are dealing here with successive-cyclic movement, there is no agreement between the IP and the CP that the IP merges with, which means that the resulting object is unlabeled (63).

(62) IP_j $ø_{say}$ [$_{CP}$ t_j kong t_j]]] (58)-(59)
(63) [$_?$ IP_j **[**$_{\mathbf{CP}}$ kong t_j]]]]

(58)-(59) are then the CP domain counterparts of (46b)/(49). (63) gives the relevant structure at the point when the IP moves, merging with CP. Since we are dealing with successive-cyclic movement, there is no agreement at the point of this merger. We then get an unlabeled object; as discussed above, since there was a further merger and the CP did not project, the phasal domain is closed with the CP sister of the IP sent to spell-out. Exactly as discussed above for (46b)/(49) regarding the vP domain, since there is only one copy of the IP in the CP spell-out domain, the PF deletion operation that deletes non-initial copies, which treats each phase as a separate unit, then does not delete it when it applies to the first spell-out domain. What PF then gets is (64), with what follows *kong* triggering tone sandhi on *kong*, *kong* not being sentence final.

(64) **[**$_{\mathbf{CP}}$ kong [A-sin si taipak lang]]]]

Notice, however, that it is crucial here that what is sent to spell-out is the whole phase. If what is sent to spell-out is a phasal complement, there would be no point of the derivation where *kong* is sent to spell-out and where *kong* is not sentence final.

Recall also that moved elements do not always license tone sandhi before movement, as shown by (61), where the object does not license tone sandhi on *kong* before moving. The contrast between (58)-(59) and (61) in fact falls out straightforwardly under the current analysis. In (61), just as in (47b)/(50), there are two copies of the DO in the first spell-out domain (see (65)). The PF deletion operation that deletes non-initial copies then deletes the lowest DO copy; *kong* being sentence-final, it doesn't undergo tone sandhi.

(65) **[**$_{\mathbf{vP}}$ DO_i [$_{vP}$ kong t_i]]

The current analysis thus captures the contrast between (58)-(59) and (61) with respect to derivational tone sandhi licensing. (58)-(59) are the counterparts of (46b)/(49) on the CP level (i.e. CP is the counterpart of vP) and (61) is the counterpart of (47b)/(50). The analysis of the English stress assignment facts from the previous section thus straightforwardly extends to the tone sandhi facts. That the analysis achieves uniformity here can be taken as an argument for it. More importantly, what we are seeing here is another case where successive-cyclic movement and final movement behave differently for the syntax-phonology interface. In both cases, SCM but not final movement shows derivational phonology licensing effects, which is captured by the proposed analysis. Additionally, we have seen here another argument for phasal spell-out (as opposed to phasal-complement spell-out).

7. Back to raddopiamento fonosintactico (RF)

Returning now to RF, a question arises whether RF can be triggered under SCM? Under the current proposal (in contrast to the standard analysis), *~~who~~* and *that* are in different spell-out domains/phases in *who do you think ~~who~~ that Mary saw*. The lack of RF in such examples in Abruzzese is then predicted. RF is indeed not possible here: 'whom' cannot trigger RF on the embedded clause complementizer *ca* in (66) (although it

[14]The sentence-final *kong* heads the complement of this null 'say'.

can trigger it on the subject, just as in (26)).

(66) Lu waglione chə ttu t'apinzə ca si viste
the boy whom you self-think that are seen (Roberta D'Alessandro, p.c.)

To summarize the discussion in the last three sections, we have seen that successive-cyclic movement and final movement to the same 'phasal edge' behave differently for the syntax-phonology interface, which was captured by the proposed analysis.

8. Further consequences

In this section I will discuss several additional syntactic consequences of the proposed analysis.

8.1. Criterial freezing

One consequence of the above analysis is that it deduces criterial freezing (Rizzi 2006). Under criterial freezing, *what* in (43) is not available for further movement, while *what* in (41) is (at the point when it is located in the embedded clause SpecCP). In the current system, *what* in (41)/(42) is not in the embedded clause spell-out domain, hence available for further movement, while *what* in (43)/(44) is in the embedded clause spell-out domain, hence unavailable for further movement, which captures the effect in question. (Notice that the PIC, which is anyway not needed under the current analysis, cannot differentiate the two constructions.)

Interestingly, this analysis does not only ban movement of the element in interrogative SpecCP, it also bans subextraction from it. Such subextraction is sometimes assumed to be possible (see e.g. Chomsky 1986, Rizzi 2006) but there is evidence that it is actually disallowed. E.g., (67a) is only slightly degraded. It is not clear whether its slightly degraded status is due to a processing effect (since the example is rather complicated) or due to a weak locality violation (D-linking in general quite often weakens locality violations). The possibilities can be teased apart with adjunct extraction: as Takahashi (1994) shows, adjunct extraction from SpecCP is completely unacceptable (see (67b)), which confirms that we are indeed dealing with a locality violation here (note also the degraded status of (68)).

(67) a. ?Which athletes$_i$ do you wonder [which friends of t$_i$] John met?
b. *How$_i$ do you wonder [how likely to fix the car t$_i$] John is (Takahashi 1994)
(68) ??/*Whose book$_i$ do you wonder [$_{CP}$ [how many reviews of t$_i$]$_j$ John read t$_j$? (Corver in press)

Torrego (1985) claims that subextraction from SpecCP is possible in Spanish based on (69a). However, Gallego (2007) demonstrates that such examples involve a prothetic object, the extracted element being an object of the higher verb, as shown in (69b). When the prothetic object option is blocked by reconstruction, such examples become unacceptable, as shown by (69c) (the same quite generally holds with verbs that disallow prothetic objects), which indicates that subextraction from interrogative SpecCP is disallowed.

(69) a. Este es el autor del que no sabemos [qué libros] leer
this is the author by whom not (we) know what books read (Chomsky 1986)
b. Este es el autor [del que]$_i$ no sabemos t$_i$ [$_{CP}$ [qué libros]$_j$ leer t$_j$]
c. *[$_{CP}$ [De qué hijo suyo$_j$]$_i$ C sabes [$_{CP}$ [qué novelas t$_i$] C ha leído todo padre$_j$]]?
of what son his know-2.SG what novels have-3.SG read every father
'which son of his do you know which novels by every father has read?' (Gallego 2007)

The proposed analysis also makes a very interesting prediction: just as there is a contrast with respect to movement from a position depending on its (non-)criterial status, there should the same kind of a contrast with respect to sub-extraction from a position depending on its (non-)criterial status. In other words, just like there is a difference with respect to movement from a criterial and non-criterial SpecCP, illustrated by the contrast between (41) and (43), all else being equal, there should be a difference with subextraction from a criterial and a non-criterial SpecCP.

Let us then consider subextraction with successive-cyclic movement. The quantifier float example from West Ulster English in (70) shows that, in contrast to subextraction from criterial SpecCP, subextraction from a non-criterial SpecCP is indeed possible.

(70) What$_i$ did he say [all t$_i$] that he wanted? (McCloskey 2000)

Particularly interesting in this respect is possessor-stranding extraction in English. Davis (2021) shows that some dialects of English allow extraction of a possessor that strands the possessive affix, as in (71). Significantly, such extraction is possible under SCM from non-interrogative SpecCP, as in (72).

(71) Mary is the author [$_{CP}$ **who**$_k$ they said [t$_k$ **'s** new book] is good]
(72) **Who**$_i$ do you think [$_{CP}$ [t$_i$**'s** cat]$_j$ she found t$_j$ today]? (Davis 2021)

Even more interestingly, such subextraction from a criterial SpecCP is disallowed, as shown by (73), as expected under the current analysis.

(73) ***Who**$_i$ do you wonder [$_{CP}$ [t$_i$**'s** friend of which athletes]$_j$ John met t$_j$]? (Colin Davis, p.c.)

One could, however, argue that the wh-possessor is the element that undergoes wh-movement here, pied-piping the larger phrase, *which athletes* being too deeply embedded to be able to do that. This could then be a case of a simple criterial freezing, where the element that undergoes (more precisely, is "responsible" for) wh-movement cannot undergo further movement. This issue is controlled for in the paradigm in (74). (74a) shows that *which* can pied-pipe the larger phrase in which it is embedded when undergoing movement to SpecCP. Note that the SpecCP in question is a criterial SpecCP, it is not an SCM SpecCP. For some speakers, including Colin Davis, extraction from a relative clause results in a very weak violation. This is responsible for the ? in (74b). Crucially, (74c) is way, way worse than (74b), so the issue with (74c) cannot simply be extraction from a relative clause.

(74) a. Syntax is a topic [[his ideas on which]$_i$ the professors used to debate about t$_i$ often]
b. ?He's the one [whose ideas on which]$_i$ syntax is a topic [the professors used to debate about t$_i$ often]
c.*He's the one **who**$_j$ syntax is a topic [t$_j$**'s** ideas about which]$_i$ the professors used to debate about t$_i$ often
(Colin Davis, p.c.)

The contrast between (72) and (74c) shows that there is a difference with subextraction from SpecCP which depends on whether the SpecCP in question is a criterial position (in (74c) it is, and in (72) it is not). This quite strongly confirms the above analysis, on which the contrast between movement from a criterial freezing SpecCP and a non-criterial freezing SpecCP is expected to be mirrored with subextraction, i.e. subextraction from criterial freezing and non-criterial freezing SpecCP. Notice also that a simple criterial freezing analysis (see Rizzi 2006) cannot get the latter contrast.

8.2. Edge of the edge

The current system has another desirable side effect: it captures Hiraiwa's (2005) observation that the edge of the edge of phase XP cannot move outside of phase XP. Thus, Hiraiwa shows that in (56), what is located in SpecXP or adjoined to XP cannot undergo movement outside of HP.[15] As an illustration, Hiraiwa observes that a possessor, which is at the edge of the DP phase, can extract out of an object in Kinyarwarnda, but a possessor of a possessor, which is at the edge of the edge of the DP phase, cannot. This is problematic for the traditional, phasal complement spell-out/PIC-based approach to phases (since edge of the edge is still at the phasal edge hence should be accessible from the outside) but falls out straightforwardly from the current system. Consider (56). Since KP does not undergo movement outside of HP, it must be undergoing feature-sharing with its its sister. This means that its sister is not sent to spell-out: what is sent to spell-out is in fact the whole HP, which makes everything within HP inaccessible from outside of HP, capturing Hiraiwa's observation, which is problematic for the alternative approach under consideration.

(56) [$_{HP}$ KP [$_{H'}$ H ...]]   (HP=phase)

8.3. Head-movement

It appears that syntactic head-movement of a phase head is blocked under the current analysis since, one way or another, the head will be in what is sent to spell-out (in contrast to the Spec). One may then think that the current analysis would require treating head-movement as a PF operation, as is in fact often done. There is, however, an interfering factor here: head-movement of a phase head can void phasehood (see e.g. Bošković 2013b, 2015, den Dikken 2007, Gallego and Uriagereka 2007; cf. especially Bošković's 2013b generalization that traces do not head islands), so the issue in question may not even arise.

To illustrate the effect in question, Bošković (2015) shows that the Complex NP Constraint is voided in Setswana (see (75)), where the noun precedes <u>all</u> DP elements, indicating N-to-D movement, and the same holds for Bulu and Swahili, also Bantu languages with N-to-D movement (see Carstens 2011, 2017 on N-to-D movement in Bantu). Bošković (2015) provides a phase-based account of the Complex NP Constraint, as well as an account of (75), whose crucial ingredient is head-movement voiding phasehood.

(75) Ke m-ang yo o utlw-ile-ng ma-gatwe a gore ntšǎ e lom-ile?
it C1-who C1Rel 2sgSM hear-Perf-Rel C6-rumor C6SM that C9-dog C9SM bite-Perf
'Who did you hear rumors that a dog bit?' (Bošković 2015)

As another illustration, Galician has D-incorporation, which involves incorporation of D into V and which quite generally voids the islandhood of the DP in question (see Uriagereka 1988, Bošković 2013b). To illustrate, extraction from adjuncts is banned in Galician (76). However, the ban is voided with D-incorporation, as in (77) (other DP island effects are also voided under D-incorporation).

(76) ??de que semana$_j$ traballastedes [$_{DP}$ o Luns t$_j$]
of which week worked the Monday
'Of which week did you guys work the Monday?'
(77) de que semana$_j$ traballastede-lo$_i$ [$_{DP}$[$_{D'}$ t$_i$ Luns t$_j$]] (Bošković 2016a)

At any rate, the point of the above discussion is that the current system does not require treating head-movement as a PF operation. While it appeared that syntactic head-movement of a phase head is blocked under the current analysis, the interfering factor is that such head-movement can actually void phasehood

[15]Note that if the element in a criterial position is a phase, subextraction from a criterial position would involve extraction of the edge of the edge under the standard phasal complement spell-out/PIC-based approach to phases. It should, however, be noted that none of Hiraiwa's cases, which motivated his restriction, involve a criterial position (Hiraiwa was concerned with extraction from SpecDP and SpecvP).

to start with, voiding any phasehood effects.[16]

There is another possibility: just like under the standard analysis movement to the edge of phase XP is assumed to take place before the complement of X is sent to spell-out, which also applies to head-movement (e.g. V-to-v movement is assumed to take place before VP is sent to spell-out), we can simply assume that in the current system, movement of X^0 to the next head up takes place before phase XP is sent to spell-out (this would basically mirror the relevant assumption regarding head-movement of the "standard" system in that in both cases, the head of what is going to be sent to spell-out moves before the relevant phrase is sent to spell-out). I will assume this option in the discussion below.

8.4. A speculation on sluicing

Sluicing, illustrated by (78), is standardly assumed to involve ellipsis of the complement of C, i.e. IP ellipsis.

(78) Mary fired someone, but I don't know who ~~Mary fired~~

A puzzle for this analysis is raised by (79). Matrix questions involve inversion. In fact, without sluicing, we get *who will Mary fire*, **who Mary will fire* being unacceptable. But the auxiliary cannot survive sluicing in (79), which is surprising if sluicing involves deletion of the complement of C, given that the auxiliary should be in C here.

(79) Mary will fire someone.
  a. Who b. *Who will

Merchant (2001) observes that there is actually a bigger issue here.[17] Overt material in C in general cannot survive sluicing. Thus, overt C is generally banned under sluicing, as illustrated below with Indonesian (without sluicing, the complementizer *yang* would be present).

(80) Ali membeli sesuatu, tapi saya tidak tahu apa (*yang)
  Ali meng-buy something, but 1s neg know what COMP (Fortin 2007)

In other words, it looks like sluicing involves C'-deletion though intermediate levels of this sort are supposed to be invisible (see e.g. Chomsky 1995). The system proposed above enables us to capture the C'-deletion effect without actual bar-level deletion. A number of authors have made various proposals involving the mechanism of marking for ellipsis, which takes place in the syntax, see Holmberg (2001), Aelbrecht (2010), Bošković (2014), Chomsky (2001), Heck and Müller (2003), Müller (2011), among others. What does that entail? I sugges that ellipsis marking blocks immediate agreement/feature-sharing. A result of this is that sluicing is treated like (41)/(42), not (43)/(44), within the current system. In other words, it is treated like successive-cyclic movement, not final movement. This means that the sister of the wh-phrase is sent to spell-out, just like the sister of *what* is sent to spell-out in (42) (which is not what

[16]Note that this can be implemented without any look-ahead involved, see Bošković (2015) and Bošković (2013b) for two different ways of doing it.

[17]Landau (2020) argues that sluicing does not involve inversion based on what would be inverted negation with sluicing undergoing scopal reconstruction, which overtly realized inverted negation does not. There is, however, an interfering factor: the overtly realized negation in the relevant examples is focused (e.g. in *don't everyone expect a raise* from Potsdam 1998), which the deleted negation in sluicing is obviously not. The interfering factor is that focus quite generally blocks scopal reconstruction (see e.g. Shibata 2015 for relevant examples from a number of languages. For the sake of a quick illustration, negation cannot scope over the subject in *Only that boy didn't come*, in contrast to *Every boy didn't come*). At any rate, the issue in question does not arise with respect to examples like (80).

happens in (44)). The sister of *what* includes the C, i.e. whatever is located in C. (Basically, we get the C'-deletion effect without actual C'-deletion since what is deleted is a full phrase, and a phase, at the point of deletion; see Messick 2014 for a different approach).

There is an exception to Merchant's generalization, involving Bulgarian enclitic C *li.* As discussed in Bošković (2001) and illustrated by (81), *li* can survive sluicing.

(81) Kogo li?
whom Q
'Whom?' (Bošković 2001)

I suggest that the clitic *li* encliticizes to the preceding element (still within its phase) hence it is not part of the CP that is elided (in other words, since *li* cliticizes to *kogo* before sluicing, it is not part of the sister of the wh-phrase that undergoes sluicing ellipsis).[18] [19]

9. Some thoughts on phases and edges

9.1. Direct vs indirect reference theories of the syntax-phonology interface

In this section I will discuss briefly the consequences of the above discussion for the debate regarding direct vs indirect reference theories of the syntax-phonology interface.

Most current direct-reference theories of the syntax-phonology interface (DR) assume that phonological interaction between X and Y is affected directly by <u>spell-out domains</u>. In indirect reference theories (IR), the interface is conditioned by prosodic constituency, which is itself taken to be determined by syntactic structure, e.g. intonational phrase (InP) is typically taken to correspond to a <u>phase</u>.

[18]Gungbe focus marker, which survives ellipsis (Liptak and Aboh 2013), may be amenable to the same analysis. For another case that looks like movement out of an ellipsis site, see Jenkins (2024), who argues that the manner adverb in (ic), which normally cannot occur in the pre-aux position (ib), moves to its surface position before ellipsis in (ic).
(i) a. John will completely lose his mind.
b. *John completely will lose his mind.
c. John partially lost his mind, and Bill completely did. (Lasnik 1998)
For another relevant case involving clitic movement, see Talić (2018).

[19]It is worth noting here that the current system can also capture Bošković's (2016c) only-the-outmost-edge-is-accessible observation. In particular, working within the standard edge-based phasal system, Bošković (2016c) shows that when a phase has multiple edges, only the outmost edge is accessible, i.e. can undergo movement. However, if that edge undergoes movement, then the next highest edge is accessible, as outlined in (i).
(i) $[_{XP}$ α $[_{XP}$ β $[_{XP}$ γ XP a phase: (a) α can move, β and γ cannot move; (b) if α is a trace β can move
(i) can be captured in the current system if we adopt the following assumptions that were independently argued for in the works cited below:
(ii) A moving element has an uninterpretable feature (Bošković 2007)
(iii) The presence of an uninterpretable feature blocks labeling (Bošković in press a)
XP in (iv) has multiple specifiers, α being the outmost Spec. If the outmost edge, α, is going to undergo movement it will have an uninterpretable feature, uK, given (ii). This will then block labeling, given (iii), as shown in (iv). Since there was a further merger and XP did not project, the sister of α, XP, is then sent to spell-out, which means that lower Specs of XP cannot undergo movement. Suppose now that both α and β will undergo movement. Then, both will have a uK. What we get then is (v), where bolded XP is sent to spell-out, capturing the effect that movement of the outmost edge enables the next highest edge to move. Importantly, if α in (v) doesn't move/doesn't have a uK and undergoes agreement, what will be spelled out is the whole phrase that dominates α (after additional merger), which means that in this scenario, β would not be able to move even with a uK. We thus capture the outmost edge effect noted in Bošković (2016c) and outlined in (i).
(iv) $[_{?}$ α (uK) $\mathbf{[_{XP}}$ β $[_{XP}$ γ sister of α sent to spell-out
(v) $[_{?}$ α (uK) $[_{?}$ β (uK) $\mathbf{[_{XP}}$ γ sister of β sent to spell-out

The data regarding (4) fit both IR and DR since the relevant locality domains are indeed defined in terms of spell-out domains, but since spell-domains are phases they also correspond to intonational phrases (InPs). If spell-out domains are mapped to InPs, it appears then that there is no difference between assuming that the locality of the relevant PF operations is stated directly in terms of spell-out domains or in terms of prosodic constituents, given that they correspond to the same spell-out domains (this is not the case if spell-out domains corresponds to phasal complements). Thus, under the current analysis it seems even harder to tease the two apart.

The IR approach, however, can handle cases where PF operations exceptionally occur across spell-out domain boundaries. I implement this (see Selkirk 2011 for an overview of other implementations) in terms of two stages of prosodic phrasing (e.g. Schütze 1994, Dresher 1994, Nespor and Vogel 1986), the initial prosodic phrasing and the readjustment stage. The initial prosodic phrasing stage corresponds to spell-out domains, with spell-out domains corresponding to InPs. However, there is also the readjustment stage, which can erase certain InP boundaries (e.g. in the case of elements that are not sufficiently long prosodically), or insert additional InP boundaries (e.g. in the case of parentheticals), and where the notion of edge, properly defined, plays a role (I return to this later).

From this perspective, Chichewa (82), where the InP spans from the beginning of the sentence to the end of the relative, is treated in terms of readjustment, which erases e.g. the InP boundary at the relative clause CP phase. (In Chichewa, the next-to-last vowel of an InP is long (there are no other long vowels), hence a long vowel indicates an InP boundary that occurs to the right of the following vowel).

(82) (Ma-kóló a-na-pátsíra [DP [CP mwaná a-méné á-ná-wa-chez**éera**]])
CL6-parent 6SUBJ-PST1-give CL1.child 1-REL 1SUBJ-PST2-6OBJ-visit
([DP ndalámá zá mú-longo w**áake**]).
CL10.money 10.of CL1-sister 1.her
'The parents gave [the child who visited them] money for her sister.' (Cheng and Downing 2016)

In summary, spell-out domains correspond to phases. Assuming that under DR, the locality of PF operations is stated directly on spell-out domains while under IR it is stated in terms of prosodic structure which is itself conditioned by syntactic structure, with InP corresponding to a phase, it is even harder to distinguish the two if spell-out domains correspond to phases. However, IR can capture exceptional cases where PF operations span spell-out domains, given that in these cases the prosodic structure does not faithfully correspond to the syntactic structure, as a result of which the locality of PF operations is not rigidly tied to spell-out domains under IR. There is also more relevant crosslinguistic variation than one would expect to find under DR (see e.g. Selkirk 2011), which IR can capture more easily through its treatment of exceptional cases through language-specific readjustment processes.

At any rate, what we are then led to is a system with two stages of prosodic mapping, the first one is spell-out domain sensitive, but then readjustments can be done. Those readjustments need to be constrained in a principled way; possibly, they can take place only if motivated.

Consider in this respect raddoppiamento sintattico (RS) in Italian, where, in a word1 word2 sequence, the initial consonant of w2 is lengthened (within a phonological phrase) if w1 ends in a vowel with the main stress of the word, and the initial consonant of w2 is followed by a non-nasal sonorant. RS is illustrated by (83).

(83) La scimmia aveva appena mangiato metá [b:]anana
the monkey had just eat.PP half banana

'The monkey had just eaten half a banana' (Nespor and Vogel 1986:38)

RS is not possible between a verb and the object:

(84) Porterá **d**ue tigri fuori dalla gabbia Relevant RS (bolded): *
take.Fut.3sg. two tigers outside from.the cage
'He will take two tigers out of the cage' (Nespor and Vogel 1986:173)

There is an exception though: when the object is non-branching (i.e. light):

(85) Se prenderá **q**ualcosa prenderá **t**ordi Relevant RS (bolded): OK
if catch.Fut.3sg. something catch.Fut.3sg. thrushes
'If he catches something, he will catch thrushes. (Nespor and Vogel 1986:172)

Inkelas and Zec (1995) argue that a preferred phonological phrase has at least two phonological words. Dobashi (2003) in turn argues that this is what is responsible for the exceptional possibility of RS in (85). Adjusting this analysis to phasal spell-out, the main verb in (84) is in T (since Italian is a V-to-T language), the object is within vP, which is a phase and a spell-out domain that would normally block RS. However, prosodically motivated readjustment takes place in (85), to get a phonological phrase that consists of at least two phonological words (for relevant discussion based on Irish, see Elfner 2012).

In the current system, where what is sent to spell-out is the full phase, it seems that theoretically, there is no concept of phasal edge for PF. This is actually not completely true. Under the current approach, there is a concept of edge at PF, but it is not there for all cases (in contrast to the phasal complement spell-out model). It is there only for some cases, where the intonational phrase boundary is readjusted: readjustment (which may require prosodic motivation) is possible only at the edge, which means for the initial/highest (see below) element following the intonational phrase boundary.

There is an interesting difference here with the standard edge-based model that involves phasal complement spell-out. Under the most natural interpretation, the notion of edge in the current system would involve linear order, not structural height: readjustment would affect the first element in the relevant domain, which does not have to be the element in the syntactic edge, i.e. Spec. Consider in this respect how this concept of edge would treat (85), in comparison to the "standard" height-based concept of the edge. The verb in (85) is in T, outside of the vP phase, this means that in both the current and the standard system, we are dealing here with phonological interaction across phasal boundaries, where the notion of edge matters. In the standard system (see Chomsky 2000), the object would need to be located in SpecvP, reflecting the syntactic notion of edgehood. This is not the case in the current system. The object merely needs to be the initial element of the vP phase (i.e. the vP phase spell-out domain), it may in fact still be located in its in situ position in VP. The former analysis requires object shift, which the current analysis does not require. I am actually not aware of any convincing arguments for object shift in such constructions in Italian, which can be taken to favor the current analysis (and the linearly based notion of the edge).[20]

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[20]The above discussion is in many respects programmatic. It also raises a number of interesting questions that cannot be discussed in the confines of this paper. E.g., is there anything like this on the LF side, i.e. are there readjustments of this type? Maybe this is the case with binding cases where an anaphor is bound "at the edge", as in *She wonders which picture of herself he sold*, given that the whole CP is a phase and a spell-out domain now (in fact, maybe logophoricity is at least partially about such readjustments; the reader is also referred to Saito (2025) for arguments based on anaphor binding that what is transferred to the semantics interface is a phase, not phasal complement.

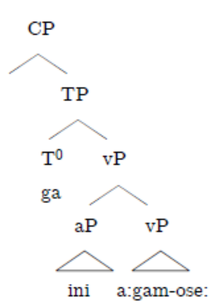
When it comes to the PF side, not all phasal boundaries are the same, some are more prone to readjustments, in fact, we have a hierarchy CP-vP-DP for the possibility of readjustment in PF (DP is often not a separate intonational

In addition, not all phonological processes may be able to trigger readjustment (this kind of selectivity, discussed directly below, also favors the current selective notion of *edge* that arises through readjustment over the “standard” edge-is-always-the-edge notion). Consider syllabification/footing and vowel hiatus resolution in Ojibwe. Ojibwe generally disallows word-medial monomoraic feet. However, it exceptionally allows degenerate feet if there is no material spelled out in the same phase for it to resyllabify with (see Newell and Piggott 2014). Consider in this respect (86), comparing it with (87) (the contrast was noted in Newell in press). What is important here is that the Tense marker *ga* in (86) is monomoraic, while the Tense marker in (87), *gi:*, is not, [i:] being a long vowel. Assume that, as argued above, what is spelled out is phases and that vP is a phase. Newell (in press) argues that the aP *ini* is adjoined to vP, as in the structure in (88). I assume that the vP in (88) is what is sent to spell-out.[21] $\text{T}_{\text{[fut]}}$, which is realized as monomoraic *ga*, is then added. *Ga* is, however, a degenerate foot. Readjustment then takes place, to allow refooting across a phasal boundary. Due to being a degenerate foot, *ga* refoots with *ini,* which is inside the vP phase (this does not happen in (87) since gi: is not a degenerate foot). Since *ga* and *ini* are now inside the same domain, vowel hiatus also needs to be resolved, which is done by the insertion of an epenthetic [d] between [*g**a***] and [***i**ni*].

(86) [ga [[ini $_{\text{aP}}$] [a:gam-ose: $_{\text{vP}}$] $_{\text{CP}}$] (gadì)(nì)(á:)(gamò)(sè:)
Future-Away-snowshoe-walk
‘He will walk there in snowshoes.’

(87) [gi: [[ini $_{\text{aP}}$] [a:gam-ose: $_{\text{vP}}$] $_{\text{CP}}$] (gì:)(inì)(á:)(gamò)(sè:)
Past-Away-snowshoe-walk
‘He walked there in snowshoes.’ (Newell in press:17)

(88)



Thus, in (86)/(88), both refooting and vowel hiatus resolution occur across elements in two distinct spell-out domains/phases, the readjustment in question being motivated by the dispreference for word-medial monomoraic feet in Ojibwe. This then triggered vowel hiatus resolution. Notice, however, that vowel hiatus resolution itself cannot trigger readjustment. If it could, readjustment should take place between [gi:] and [ini] in (87) to trigger vowel hiatus resolution across spell-out domains. But this does not happen.[22] This

phrase (InP), which is not surprising given that it is phonologically the lightest). Alternatively, when it comes to syntax-prosody correspondence, it is possible that each phase must correspond to some prosodic constituent, not necessarily an InP (this would be less different from Selkirk’s 2006, 2009 Match Theory. Prosodic weight obviously matters here, see Selkirk 2011 for other factors, cf. also the Strict Layer Hypothesis—under the hypothesis, there cannot be an InP immediately dominating an InP, which can often prevent DP from being parsed as an InP (even though it is a phase), e.g. if a vP is parsed as an InP, the DP in SpecvP cannot be.)

[21] I depart from Newell (in press) in this respect. The analysis given below actually differs significantly from Newell’s.

[22] Apparently, vowel hiatus resolution cannot be triggered by what Newell (in press) calls aP (the same actually holds for refooting, though the relevant data are not discussed above).

then indicates that the readjustment we are discussing here is not triggered by all phonological processes, which allows us to capture not just potential crosslinguistic variation (of the relevant sort), but also variation within a single language, like in this case the different behavior of the constraint against word-medial monomoraic feet and vowel hiatus resolution. As noted above, this kind of selectivity also favors the current selective notion of *edge* that arises through readjustment over the "standard" edge-is-always-the-edge approach, where it is difficult to capture the different behavior of (86) and (87) regarding vowel hiatus resolution (cf. also (i) in fn 22).

At any rate, the reader should bear in mind that much of this paper is programmatic, which means that the emphasis is not on what the system cannot do, but on what it can do, especially where it differs from the more standard system in this respect, i.e. where it provides fresh perspectives.[23]

10. Conclusion

In conclusion, evidence from syntax-phonology interaction quite clearly points to the conclusion that what is sent to spell-out is phases, not phasal complements. This, however, raises an issue for the interaction of successive-cyclic movement and spell-out: if what is sent to spell-out is a full phase and inaccessible to syntax, if successive-cyclic movement were to target phases, i.e. phasal edges, as is standardly assumed, an element undergoing movement would get frozen with the first step of successive-cyclic movement since it would be part of a spelled-out unit, which would make it inaccessible. I have shown that the contradiction is resolved under the labeling theory: spell-out can correspond to a phase and successive-cyclic movement can still target a phase. Importantly, phase XP with YP in its Spec does not always behave in the same way for spell-out (indicating contextuality of spell-out—see Bošković 2024, in press b for a broader discussion of contextuality): with successive-cyclic movement (see (38)-(39)), the sister of YP is sent to spell-out, otherwise (see (43)-(44)) the full XP is sent to spell-out, although both cases involve phasal spell-out. The system was shown to resolve a number of outstanding issues, e.g. certain stress assignment puzzles in English, tone sandhi issues in Taiwanese, and the seeming C'-deletion with sluicing. Additionally, it deduces criterial freezing in a way that also captures its parallel with subextraction, also deducing the edge-of-the-edge restriction. The discussion in the paper also favors indirect reference over direct reference theories of the syntax-phonology interface, with the notion of readjustment at the interface that involves a selective and linearly-based notion of edge. The paper also shows that successive-cyclic movement and final movement to the same position can behave differently for the syntax-phonology interface, i.e. they can have different reflexes for phonology.

References

Note that that vowel hiatus is resolved between elements within a phase is confirmed by (i), where the vowel hiatus between the final vowel of the root and the initial vowel of the diminutive suffix (they are spelled out in the same phase) is resolved by deletion of the second vowel (after a consonant, the diminutive suffix is realized as [e:ns]).

(i) ase:ma:-e:ns = ase:ma:ns
tobacco-DIM 'cigarette' (Piggott and Newell 2014: 337)

[23] I will not discuss here Agree, given that there is some controversy regarding whether Agree is subject to the PIC/phase-constrained, with Bošković (2007) giving a number of empirical arguments that it is not. If it is subject to it, in the current system it would be phase constrained. However, a number of authors (for relevant discussion, see e.g. Arregi and Nevins 2012, Benmamoun, Bhatia and Polinsky 2009, Bhatt and Walkow 2013, Bobaljik 2008) have argued that Agree (or some aspects of it) may take place in PF, in which case it could be subject to the edge-based readjustment discussed above.

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